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from keras.utils import plot_model
from keras.models import Model
from keras.layers import Input
from keras.layers import Dense
from keras.layers.pooling import MaxPooling2D
from keras.layers.convolutional import Conv2D

visible = Input(shape=(64, 64, 1))
conv1 = Conv2D(32, kernel_size=(3, 3), activation='relu')(visible)
pool1 = MaxPooling2D(pool_size=(2, 2))(conv1)
conv2 = Conv2D(16, kernel_size=(2, 2))(pool1)
pool2 = MaxPooling2D(pool_size=(2, 2))(conv2)
hidden1 = Dense(20, activation='sigmoid')(pool2)
output = Dense(1, activation='relu')(hidden1)
model = Model(inputs=visible, outputs=output)

model.summary()

# plot graph
plot_model(model, to_file='convolutional_neural_network.png')

```